CHANGELOG

Changelog

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

[Unreleased]

Added

- **Wave 94 — autonomous completion loop (subagent-driven): defect fixes + concurrency hardening + unfinished-work report (conductor-gated, per-package `-race`):**
  - **HXC-1637:** `make build` now compiles every real `./cmd/*` Go binary into `bin/` (was a non-existent `./cmd/helix-cluster`); the `zig`/`cmake` steps are guarded by `command -v` so a missing native toolchain no longer hard-fails the build. Verified `go build -o bin/ ./cmd/...` (22 binaries).
  - **HXC-1638:** new real `cmd/helixctl` — a cobra gRPC **client** for the BuildService (`build submit/status/cancel/logs`) with in-process round-trip tests under `-race` (no PASS-bluff: `list` omitted because no `ListBuilds` RPC exists). `docs/guides/getting-started.md` + `docs/USER_MANUAL.md` corrected to the real command surface.
  - **Concurrency hardening (audit findings F1–F7), each TDD + `go test -race` green:** SWIM `Stop()`/`Leave()` double-close panic → `sync.Once`-guarded teardown (F1); SWIM `HealthyMembers` data race on `Member.State` → read via `m.IsHealthy()` (F2); SWIM probe goroutine now ctx-aware + `wg`-tracked (F3); `FailureDetector.Confirm/Refute` no longer hold `fd.mu` across the protocol callback — fixes the `fd.mu→p.mu` lock-order self-deadlock (F4); native session PTY double-close / use-after-close → close-once under `sess.mu` + I/O serialized against close (F5); scheduler `placements` map unbounded growth → `Complete`/`Release` completion path (F6); EventBus `PublishAsync` unbounded goroutines → bounded dispatcher + non-blocking `trySend` fast path (F7/F10, EventBus submodule).
  - **`docs/UNFINISHED_WORK_REPORT.md`** — comprehensive grounded inventory + phased completion plan: dead-code/orphaned-feature finding (178 of 212 `pkg/` packages unwired, ~51.4k LOC; `helixd` a hollow shell), coverage gaps, Snyk/SonarQube-via-Podman plan, registry Bucket A/B/C split, PRR gaps, and the owner-decision gates.

- **Documentation completeness pass — full operator+user docs, SQL schema, diagrams, test-coverage report (all md+html+pdf+docx, docs_chain-tracked):**
  - `docs/USER_MANUAL.md` (operator), `docs/USER_GUIDE.md` (end-user), `docs/DATABASE_SCHEMA.md` (every postgres table/column/index/trigger + Mermaid ER + migrate-chain-vs-primary-schema reconciliation note), `docs/ARCHITECTURE_DIAGRAMS.md` (Mermaid: L0-L7, control-plane, request flow, tier matrix), `docs/TEST_COVERAGE_REPORT.md` (real measured coverage: main 82.4%/255 pkgs, security 87.8%, api/v1 14.7%; per-test-type inventory). All grounded in real code (adversarial-reviewed, no PASS-bluffs) and linked from the README documentation map; exported to HTML/PDF/DOCX via docs_chain.
  - Grounded authoring surfaced 3 real defects, now ticketed: HXC-1637 (make build targets non-existent ./cmd/helix-cluster), HXC-1638 (helixctl referenced but absent), HXC-1639 (0001_primary_schema.sql diverges from the migrate chain 001-015).
- **Wave 93 — local dependency-update maintenance (HXC-1636):** `make deps-update` (scripts/deps-update.sh) runs per-module `go get -u` + `go work sync` + `govulncheck` + `make sbom` — the no-CI local equivalent of dependabot (operator-chosen). Closes PRR item 80; PRR now 66/80 (82.5%).
- **Wave 92 — last CLAUDE-2 hotspots + ChaCha + SBOM (HXC-1633/1634/1635/1561; operator-authorized local deps; conductor-gated):**
  - HXC-1633 (CLAUDE-2): real userspace WireGuard on macOS via gVisor netstack (pkg/wireguard/netstack_darwin.go) — real Noise handshake + encrypted TCP datapath between two peers, no root; linux wgctrl path intact; go.mod records gvisor (standalone build green). Mutation neutering AddPeer fails the tunnel test.
  - HXC-1634 (CLAUDE-2): real darwin GPU probe via system_profiler -json (pkg/device) — real-host oracle match (Apple M3 Pro/19.3GB/metal). Mutation breaking JSON tags => 0 GPUs.
  - HXC-1635: make sbom via cyclonedx-gomod => CycloneDX SBOMs main/api-v1/security (96/6/2 comps), jq-validated.
  - HXC-1561 (security submodule): ChaCha20-Poly1305 AEAD suite via x/crypto (RFC-8439 KAT); AES-GCM default preserved; dep clause amended. PRR (HXC-1286) rises to 65/80 (81.25%): items 64/65/66/67 -> PASS, item 80 -> PARTIAL.
- **Wave 90 — clear ALL govulncheck-reachable advisories (HXC-1631 + HXC-1632; conductor-gated security remediation):**
  - HXC-1632: `golang.org/x/net` v0.54.0 -> v0.55.0 (clears GO-2026-5026 idna, reachable in the gateway).
  - HXC-1631: pinned `toolchain go1.26.4` in main/api/v1/security (+go.work) — clears 9 reachable Go stdlib advisories (html/template XSS, net/http HTTP/2 SETTINGS infinite-loop DoS, crypto/x509, net/mail quadratic, net Dial panic), all fixed in go1.26.3/1.26.4. Verified: `govulncheck ./...` = "No vulnerabilities found" across all 3 modules; whole-tree build + tests green under go1.26.4. Raises PRR item 34 to PARTIAL (scan run + vulns fixed; SBOM/dependabot pending). (security toolchain pin: submodule 6154969.)
- **Wave 89 — PRR-gap closures (HXC-1629 migrate wiring + HXC-1630 govulncheck), conductor-gated:**
  - `Makefile` (HXC-1629) — migrate-up/migrate-down now invoke scripts/run-migrations.sh (honor DATABASE_URL, no hardcoded secret); verified against a REAL podman postgres (15 migrations applied ->

schema_migrations v15/22 tables; down rolled back exactly 1). Raises PRR item 46 to PASS (honest score 60->61/80).
  - govulncheck (HXC-1630) — installed + scanned main/api/v1/security; 10 reachable advisories captured (evidence qa-results/security/): 9 stdlib (toolchain bump -> HXC-1631) + 1 third-party golang.org/x/net (-> HXC-1632). No auto go.mod churn; remediation tracked as tickets.
- **Wave 88 — buf-lint clean + Production Readiness Review (HXC-1628 done; HXC-1286 executed, stays open):**
  - `api/v1/buf.yaml` (HXC-1628) — `buf lint` now exits clean: 5 STYLE/naming rules excepted with per-rule rationale (proto-at-module-root layout; domain-named RPC request/response types incl. the shared KernelRequest; GPUProvider service name), DEFAULT->STANDARD category. `buf build` still passes; removing the except block re-fires 25 violations.
  - `docs/PRODUCTION_READINESS_REVIEW.md` (HXC-1286) — an 80-item production-readiness checklist with an HONEST assessment: **60/80 PASS (75%)**, below the 95% closure bar, so the ticket stays Queued with its gap-list (re-enable CI gates, macOS wireguard-go userspace, govulncheck/SBOM, Makefile migrate wiring, mTLS e2e evidence, darwin GPU probes, covgate enforcement, HelixQA challenge evidence). PASS rows cite real files/tests; NOT-READY rows cite the specific gap. docs_chain-tracked.
- **Wave 87 — api/v1 proto schema extensions (HXC-1293/1165/1294/1295/1187/1527; buf-regenerated, conductor-gated, backward-compatible):**
  - `api/v1` — Node +tier/trust_level/compute_class/arch + nested NodeThermal (HXC-1293, absorbing HXC-1165 trust_level+thermal); NodeResources +npu_tops/fpga_logic_elements/tflops_gpu (HXC-1294); ScheduleJobRequest +allowed_tiers/min_tier/power_budget_watts (HXC-1295); new `edge.proto` EdgeWorkUnit/EdgeWorkResult (HXC-1187); new `gpu_proxy.proto` GPUProvider gRPC service (DispatchKernel + DispatchKernelStream, KernelRequest/Response/Chunk) (HXC-1527). All new field numbers appended (nothing renumbered/removed); 9 round-trip tests prove field preservation; field-tag mutations fail them; buf generate idempotent; whole-tree build + consumer cmd tests green.
- **Wave 86 — proto service stubs build + generate (HXC-1117; buf now installed; conductor-gated):**
  - `api/v1` — the proto module was broken (buf.yaml mis-placed at `api/` so bare `import "types.proto"` could not resolve; `api/buf.gen.yaml` malformed for buf v1.70). Relocated the buf module root to `api/v1/` (matching `api/v1/go.mod`), removed the malformed gen config, and regenerated all 7 services' Go stubs (node/session/scheduler/health/build/advisory/security) with `protoc-gen-go/-go-grpc`. Idempotent; package stays `helixv1`. `api/v1` module + whole-tree build + consumer cmd tests green; `TestCreateSessionRequest_RoundTrip` proves marshal/unmarshal preserves all fields (mutation corrupting a field tag fails it). buf-lint STYLE conformance tracked HXC-1628 (HXC-1117).
- **Wave 85 — streaming SSE E2EE decryption (HXC-1565; security submodule `16ae574`; conductor-gated -race + tamper/order/wrong-session mutation):**
  - `security/pkg/e2ee` — `StreamDecryptor` (e2e_init + per-chunk) reassembles an ordered sequence of independently-sealed SSE chunk records against the per-request handshake-derived `Session`; any tampered, out-of-order/replayed, or wrong-session chunk POISONS the stream (AEAD tag + counter-nonce ordering), so a broken chunk cannot silently

corrupt the reassembled completion. Composed from `Session.Open`; no hand-rolled crypto. Live-relay capture deferred. Sibling HXC-1561 (ChaCha20-Poly1305) is BLOCKED pending authorization to add `golang.org/x/crypto` to the security module (HXC-1565).

- **Wave 84 — per-request response keypair (HXC-1563; security submodule `f09227f`; conductor-gated -race + isolation mutation):**
  - `security/pkg/e2ee` — `ResponseKeypair` (fresh ephemeral ML-KEM-768 Initiator per request) + worker `SealResponse` + client `OpenResponse` + `Fingerprint`, composed from existing handshake primitives (no hand-rolled crypto). A response sealed for request-1 is cryptographically unopenable with request-2's key; per-request rotation proven via distinct fingerprints. Mutation reusing one keypair fails the isolation test (HXC-1563).
- **Wave 83 — typed Avro pub/sub for the remaining domain events (HXC-1627; conductor-gated build/vet/-race + REAL-broker integration + load-bearing mutation):**
  - `pkg/events` — `NATSBackend` gains typed `Publish*`/`Subscribe*` methods for `SessionEvent`/`SchedulerEvent`/`AuditEvent` at parity with `NodeEvent`: in `WireAvro` mode each carries an Avro single-object payload (`0xC3 0x01` + fingerprint) decoded via the `SchemaRegistry`; JSON stays default. `TestNATSBackend_TypedAvroWire_RealRoundTrip` (`//go:build integration`) boots a real NATS broker via `brokertest.StartNATS` and asserts the on-wire Avro marker + exact round-trip for all three types. Mutation forcing JSON on `PublishSessionEvent` fails the on-wire assertion. All four domain events now route over the live bus in Avro mode (HXC-1627).
- **Wave 82 — Avro codec wired into the live NATS event bus (HXC-1626; conductor-gated build/vet/-race + REAL-broker integration + load-bearing mutation):**
  - `pkg/events` — `NATSBackend` gains an Avro wire mode: `PublishNodeEvent` emits an Avro single-object payload (`0xC3 0x01` + CRC-64-AVRO fingerprint) when `WireAvro` is selected and `SubscribeNodeEvents` decodes via a `SchemaRegistry` (validated routing); JSON stays the default (back-compat). Concrete Avro schemas + `Encode*`/`Decode*` helpers added for `SessionEvent`/`SchedulerEvent`/`AuditEvent` (backend typed pub/sub for those three tracked HXC-1627). Proven by `TestNATSBackend_AvroWire_RealRoundTrip` (`//go:build integration`) which boots a **real NATS+JetStream broker** via `brokertest.StartNATS`, captures the on-wire bytes and asserts the Avro marker (JSON would start with `{`), and round-trips schema evolution both directions across the live bus. Mutation forcing JSON on the publish path fails the on-wire assertion (HXC-1626).
- **Wave 81 — E2EE framed-transport concurrency + oversize hardening (HXC-1567; conductor-gated -race + load-bearing mutation):**
  - **`security` submodule (`pkg/e2ee`, `29a6193`)** — the length-prefixed framed `Transport` (`WriteRecord`/`ReadRecord` over `io.ReadWriter`) and its `DefaultMaxRecordSize` guard already existed; added `TestTransportConcurrentRoundTripWithOversizeInjection` closing the closure's composite gap — 16 concurrent independent framed round-trips stay intact while an injected 5 MiB length prefix (cap 1 KiB) is rejected with `ErrRecordTooLarge`, all under `-race`. Mutation dropping the `ReadRecord` oversize guard fails the test (HXC-1567).

- **Wave 80 — Avro event schemas + schema-validated routing (HXC-1119; careful single-threaded TDD build, conductor-gated build/vet/-race + 2 load-bearing mutation bites):**
  - `pkg/events` — a stdlib-only Avro-subset codec: flat primitive records, spec-faithful binary encoding (zig-zag varint int/long, length-prefixed string/bytes, LE-IEEE-754 double), **single-object encoding** framing (`0xC3 0x01` + 8-byte CRC-64-AVRO fingerprint) so each payload is self-describing, a `SchemaRegistry` giving **schema-validated routing** (a payload whose writer fingerprint is unregistered is rejected, never mis-decoded), and **writer→reader schema resolution** for backward+forward-compatible evolution (reader-only fields fill from defaults; writer-only fields skip; a reader-only field without a default is an unresolvable-pair error). `NodeEventSchemaV1→V2` adds a defaulted `action` field; proven both directions — a V1 consumer reads a V2 payload (skips `action`) and a V2 consumer reads a V1 payload (synthesizes `action` from default) — plus concurrent round-trip under `-race`. Live NATS-backend wiring + the other event-type schemas tracked HXC-1626 (HXC-1119).
- **Wave 79 — 2 bounded host-closeable fixes (conductor-gated build/vet/-race + load-bearing mutation bite):**
  - `pkg/multiraft` — the durability persist-error is now observable through a PUBLIC seam: a registerable `OnPersistError` hook (+ `LastPersistError` accessor) fires with the shard id + error from the same Ready-loop path that parks the un-persisted `Ready` and skips `Advance` (HXC-917); previously the fault was only `log.Printf`'d. Nil hook = prior behavior (HXC-927).
  - `pkg/cloudspot` — the GCP and Azure preemption pollers now honor the injected `WithClock` when computing `PreemptionNotice.Deadline` (both used `time.Now()` directly, silently dropping the clock; only the AWS poller honored it), making all three providers deterministically testable with a fake clock (HXC-1616).
- **Wave 78 — 4 disjoint host-closeable streams + a cross-platform SELinux fix (all conductor-gated build/vet/-race + load-bearing mutation bite, not agent self-reports):**
  - `internal/console` — fixture-driven device-classification hardening: `testdata/` `/proc`+`device-tree` trees for 6 device classes (PS4/PS5/x86-server/RK3588/Raspberry-Pi/generic) with exact-label table assertions; mutating the PS5 zen-2 SoC-match row flips `ps5_zen2` and fails the test. Non-console classes still classify `Unknown` today (positive labels tracked HXC-1624) (HXC-1158).
  - `scripts/cross-compile-agent.sh` + `Makefile` `cross-agent` target — reproducible ARM64 cross-compilation of `cmd/helix-agent`: emits linux/arm64 (ELF aarch64), linux/armv7 (ELF 32-bit ARM EABI5) and android/arm64 artifacts with a build-UUID via `-ldflags` and a `--version` print path; `crosscompile_test.go` asserts `elf.EM_AARCH64` via `debug/elf`. CGO-enabled cross (no `zig cc`/`aarch64-gnu-gcc` on host) and on-real-SBC execution are honest SKIPs (HXC-1622/HXC-1623) (HXC-1167).
  - `docs/architecture/PHASE_8C_INTEGRATION.md` — code-grounded Phase-8C integration map (attestation→scheduler, e2ee→orchestrator, marketplace) citing real `pkg/resources` symbols; scheduler attestation-wiring labeled PLANNED (verified: no `SetAttested` callers) (HXC-1614).
  - `docs/PHASE_8C_EXIT_GATE_EVIDENCE.md` — CLAUDE-1 exit-gate evidence matrix (PROVEN 9 / PARTIAL 6 / NOT-YET 7); PROVEN rows cite real `pkg/metering`+`pkg/auditproof` tests, NOT-YET rows cite Queued tickets (HXC-1613).

    - `containers` **submodule (`pkg/crossbuild`)** — SELinux bind-mount relabel made conditional + cross-platform: `:Z` private / `:z` shared **only when SELinux is active**, empty on macOS / non-SELinux Linux (was hardcoded `:Z`, breaking podman-machine/Docker-Desktop & shared-volume reuse); injectable detection, panic-recovered fail-safe, stress+chaos coverage; CLAUDE-2 §11.4.81 (HXC-1621, submodule `1598f28`).
- **Waves 76+77 — two concurrent subagent waves, 8 disjoint host-closeable streams (all independently gated build/vet/-race + load-bearing mutation bite per stream):**
    - `internal/llm` — advisory engine (MIGRATION/SCALING/CONFIG/ALERT/OPTIMIZATION) carrying rationale + confidence + risk, `PENDING` persistence, a non-LOW-risk approval gate (never auto-applied without explicit approval), and a mandatory `LLMsVerifier` that rejects hallucinated/unsafe actions; the LLM is an injectable `Advisor` so the gate is deterministic and host-closeable (HXC-1126).
    - `pkg/benchmark` — micro-benchmarker producing a real, repeatable normalized on-host CPU score (+ typed GPU TFLOPS / NPU TOPS seams), repeatability asserted within a coefficient-of-variation tolerance band (HXC-1308).
    - `pkg/security/e2ee` (**security submodule**) — gzip-before-encrypt payload compression matching the Chutes envelope framing: measurable wire-size reduction, byte-exact decrypt+gunzip round-trip, only applied when it actually shrinks (HXC-1564).
    - `cmd/dst-sim` — 1000-seed deterministic-simulation gate composing `pkg/porcupine`; exits non-zero naming the offending seed on a linearizability violation (ran 1000 seeds exit 0, deterministic across reruns) (HXC-915).
    - `docs/MVP_ARCHITECTURE.md` — living seven-layer architecture + service-communication Mermaid grounded in real packages, with a Go drift-validator that fails the build if the doc names a path that no longer exists; docs_chain-tracked (HXC-1145).
    - `pkg/gpu` — thread-safe `ProviderAdapter` registration hooks (`Register`/`List`/`Get`/`Lookup`) exposing a pool-facing listing, duplicate-name guarded, race-free under `-race` (HXC-1533).
    - `pkg/cloudspot` — real AWS IMDSv2 (token `PUT` → authenticated `GET` spot instance-action), Azure scheduled-events, and GCP preemption pollers behind `SignalSource`, with `drain→checkpoint→upload` via an injectable sink; proven against `httptest` fixtures replaying the genuine wire protocol (live cloud deferred, HXC-1617) (HXC-1328).
    - `pkg/tierdetect` — real cross-platform host-capability detection (darwin `sysctl`/arch + linux `/dev/kvm`,`/proc`,`binfmt` behind one build-tagged interface) gating tier requests pre-provision with a typed `MissingCapabilityError` (HXC-1230).
- **Wave 75 — provider hardening, security & boot readiness (5 items, all independently gated):**
    - `pkg/provider/runpod` — `Provision` no longer holds the mutex across the `ColdProvision` network RPC (concurrent cold provisions overlap; `reserved` counter preserves capacity), and `EndpointOf(id)` surfaces the reachable worker endpoint (HXC-919).
    - `pkg/marketplaceadapter` (**security**) — `buildSDL` YAML-encodes caller-controlled fields to contain manifest injection, and the Akash reputation gate is no longer bypassable by a zero-value adapter (HXC-918).
    - `pkg/provider/chutes` — `parseRetryAfter` HTTP-date branch + past-date clamp now covered by tests (HXC-920).

    - `internal/console` — `BootCoordinator` wired into the real `Registrar` via a `ReadyGate` so node registration blocks on boot readiness (HXC-1147).
    - `docs/guides/phase_02_architecture.md` — Phase-02 architecture guide + Mermaid diagram, docs_chain-tracked (HXC-1164).
- **Wave 74 — accelerator probing, power-aware edge & raft durability (5 items, all independently gated build/vet/-race + load-bearing mutation bite):**
    - `pkg/resources` accelerator reporting: `GPUInfo.TFLOPS` + `Accelerators{NPUTops,FPGALogicElements,QPUPresent}`, real per-OS probe (darwin Apple-GPU TFLOPS from `system_profiler`, linux sysfs PCI catalog) + operator override-label fallback (HXC-1300).
    - `pkg/powergater` — charging-gated `CanAcceptWork() (bool, reason)` work-acceptance guard with real darwin `pmset` / linux sysfs `PowerSource` (HXC-1175).
    - `pkg/edgeheartbeat` — battery/charging/thermal/network heartbeat + churn-timeout collector over a real loopback transport, real per-OS `TelemetrySource` (HXC-1185).
    - `pkg/multiraft` **durability fix**: a `ShardStorage` persistence error now **parks** the un-persisted `Ready` and skips `Advance` (no false "durably handled"), instead of swallowing the error (HXC-917).
    - `docs/NODE_PROVISIONING_BOUNDARY.md` — normative no-jailbreak/no-root/no-unlock provisioning boundary, registered in docs_chain (HXC-1146).
- **Foundation distributed-systems library (`pkg/`)** — a large, growing pure-Go, deterministic, mutation-tested package set. Recent additions include:
    - Consensus/coordination: `voting`, `failconfirm`, `heartbeatcoalescer`, `splitbrainalert`.
    - Replication/state: `mvcc` (B-tree time-travel store), `antientropy` (hinted-handoff + read-repair + Merkle diff), `watchmanager` (synced/unsynced/victim watch groups), `crdt/merkle`.
    - Scheduling/placement: `constraints` (Pacemaker 4-type), `preempt` (value-multiplier), `priorityqueue` (multifactor aging), `workclaim` (SKIP-LOCKED), `admissioncontrol` (N+K reserve), `budgetcap`, `qos`, `suitability`, `ewmarank`, `fallbackchain`.
    - GPU/resource & cost: `costsched`, `latencysched`, `healthmonitor`, `gpuattest` (attestation crypto: challenge/response, proof-of-GPU-work, O(1) spot-check, device-sealing), `attestadmit`, `quantization`, `local` (TCO), `pool`, `burst`.
    - Messaging/flow & edge: `flowcontrol` (K8s APF), `idempotent` (exactly-once), `rebalance` (cooperative-sticky), `informer` (list-watch cache), `redundantexec` (BOINC trust), `edgeregistry`, `edgeverify`, `slotmigration`, `hashslot`, `healthprobe`.
    - Testing/simulation: `timefault` (clock skew/freeze/monotonic injectors), BUGGIFY in `testing/dst`, `phase7matrix` (gap-matrix verifier).
    - This session (34 pure-Go units):
        - `bursthysteresis` — `BurstController` with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band (HXC-1504).
        - `providerchain` — ordered multi-tier provider fallback cascade with retriable/terminal error classification and injected-clock failover SLA (HXC-1508).

- `gpucatalog` — `SUPPORTED_GPUS` compute-multiplier catalog, `LookupMultiplier`, and attestation-aware `Score` (attested above un-attested) (HXC-1592).
- `gravaladmit` — HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and `graval.verified` label (HXC-1523).
- `modelrouter` — strategy-to-default-model resolution (latency/throughput/quality/cost) with a typed unknown-strategy error (HXC-1430).
- `gravalverify` — GraVal GPU attestation with a VRAM-ratio `>= 0.95` gate and a concurrent, race-clean `BatchVerify` pass-rate KPI (HXC-1435, HXC-1436).
- `gepetto` — HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict `>0.80` high-water anti-starvation zero-capacity cut-off (HXC-1446).
- `chutesaccount` — Chutes API model-list (TEE/price fields) and `/users/me` account-balance client over HTTP with `Authorization: Bearer` (HXC-1429).
- `cmd/burst-controller` — utilization-driven binary driving `bursthysteresis`; emits a "burst allocation request" on SPILL and a RECOVER line, runID-stamped (HXC-1531).
- `tieredcache` — hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-demotion + promotion, and per-tier hit stats (HXC-1418).
- `smartrouter` — intelligent ModelRouter (latency/throughput/cost/TEE/balanced) that always excludes unhealthy models, with a typed no-eligible-model error (HXC-1539).
- `epochresolve` — config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence) (HXC-1397).
- `balancemonitor` — USD account-balance monitor with a strict low-balance floor warning, over HTTP with `Authorization: Bearer` (HXC-1512).
- `scan` — Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock (HXC-1403).
- `llmfailover` — error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract (HXC-1600).
- `llmadapter` — Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order (HXC-1599).
- `deltacrdt` — standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge (HXC-1409).
- `gputopo` — topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe (HXC-1406).
- `fsresidency` — filesystem residency challenge: per-byte-range `ReadAt` + SHA-256 verification proving a model file is really present on disk; truncated/missing/mismatch all fail (HXC-1572).
- `economics` — `RewardDistributor` multi-token distribution with treasury/reinvest splits and a conservation invariant (`sum(ledger)+treasury+reinvest == total`), plus `GetParticipantROI` (ROI%

+ break-even days, typed unknown-participant error) (HXC-1460, HXC-1461).

- `revenueopt` — `RevenueOptimizer.OptimizeAllocation` greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias; self-contained types (HXC-1456).
- `exportcontrol` — export-control / country-code KYC gate: a controlled GPU (`H100`/`A100`) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with `ErrExportDenied`; Tier-1 is allowed (HXC-1482).
- `devicecatalog` — machine-readable 64-device taxonomy with case-insensitive substring `Lookup` resolving a discovered CPU model (e.g. `Rockchip RK3588`) to its tier / trust-level / compute-class (HXC-1331).
- `compliancedoc` — EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records, with a SHA-256 provenance hash bound to the actual record content (HXC-1481).
- `hybridkex` — hybrid post-quantum key exchange combining X25519 (`crypto/ecdh`) and ML-KEM-768 (`crypto/mlkem`) shared secrets via HKDF-SHA256 (group `X25519MLKEM768`); a tampered ML-KEM ciphertext yields a non-matching key (HXC-1476).
- `carbonsched` — carbon-aware job placement: chooses the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering (HXC-1480).
- `gpuattest` (additive `multigpu.go`) — multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors (HXC-1573).
- `workloadrouter` — `UnifiedManager.RouteWorkload`: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier that re-ranks confidential offers when the workload requires a TEE (HXC-1455).
- `metrics` (additive `tiermetrics.go`) — `TierMetrics` exposes `gpu_tier_utilization`, `gpu_cost_per_hour`, and `provider_health` Prometheus-text series with deterministic ordering and concurrent-safe setters (HXC-1535).
- `cmd/gpu-pool-manager` — GPU pool manager HTTP front-end: `POST /allocate` returns the tier-correct (lowest-tier healthy, name tiebreak) provider as JSON (`503` when none healthy) and `GET /providers` lists providers with health; httptest-gated, self-contained (HXC-1530).
- `marketplaceadapter` — `MarketplaceAdapter` interface (`Name`/`GetCurrentPricing`/`SubmitWork`) with a `Name()`-keyed dispatch `Registry` and a `ChutesAdapter` over HTTP (httptest fixture, non-2xx rejected) (HXC-1454).
- `cmd/e2ee-proxy` — transparent ML-KEM-768 (`crypto/mlkem`) + AES-256-GCM encrypting proxy: `EncryptingTransport` seals request bodies and `DecryptingHandler` opens them, so the on-wire payload between proxy and upstream is ciphertext-only while the client receives the correct plaintext; tampered ciphertext is rejected (HXC-1532).
- `e2eebench` — `MeasureHandshakeLatency` benchmarks the `hybridkex` full handshake (median + P95), proving median < 1ms on

host and enforcing per-iteration key agreement (`ErrKeyDisagreement`) (HXC-1556).
    - `archlint` + `docs/ARCHITECTURE.md` — the hardened L0–L7 architecture diagram and component map, mechanically enforced: `archlint` parses the component-map table and fails the build if any documented component maps to a package path that does not exist on disk (HXC-1424).
- Wave 67 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
    - `multiraft` — `MultiRaftManager` partitioning cluster state into independent per-shard `go.etcd.io/raft/v3` groups, each electing its own leader and committing independently via per-shard `Propose` routing over an in-process `RaftTransport`; write throughput scales with shard count and a shard re-elects on leader loss while others keep committing (HXC-1387).
    - `stonith` — STONITH fencing: a `FencingAgent` interface with IPMI / AWS-EC2 / Azure-ARM / SBD-shared-disk / NoOp drivers and a `MultiLevelFencer` multi-level fallback, with sink-side fence confirmation; the IPMI real-binary path is an honest SKIP-with-reason when `ipmitool` is absent (HXC-1401).
    - `deviceplugin` — Nomad/K8s-style gRPC device-plugin / GRES fingerprinting framework over real gRPC: plugins register full device fingerprints, the registry parses GRES descriptors, atomically applies fingerprints, allocates by request, and rejects oversubscription (HXC-1407).
    - `internal/gpu` (additive `manager_helixpow.go`) — dual-workload capacity reservation: `ReserveForHelixPoW(fraction)` + `ChutesCapacity()` reporting only the non-reserved remainder, with a strict `>0.80` Helix-load Gepetto starvation guard that zeroes Chutes capacity (HXC-1447).
    - `chutes` (additive `config.go`) — `ChutesMinerConfig` and `ValidatorConfig` deployment descriptors with `Validate()` invariants (non-empty IDs/hotkeys, `GPUCount>0`, non-negative cost, positive cache size, TEE consistency) (HXC-1439).
- Wave 68 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 69):
    - `dst` — standalone FoundationDB-style deterministic simulation harness: single-threaded seeded-RNG event loop with injectable network/disk/logical-clock; same seed => byte-for-byte identical trace + exact failure replay; 1000+ seeded sims (HXC-1419).
    - `porcupine` — self-contained WGL linearizability checker + concurrent history recorder: PASS on linearizable, FAIL (with violating op) on a seeded non-linearizable history (HXC-1421).
    - `kraft` — KRaft-style self-managed Raft metadata quorum (`go.etcd.io/raft/v3`, no ZooKeeper): replicated `CreateTopic`/ `ListTopics` + partition-leader assignment, controller election + controller-loss failover (HXC-1417).
    - `internal/chaos` — fault-injection library (PodKill/NetworkPartition/ DiskStall/ClockSkew) + canary-guarded runner with automatic rollback on health<SLO; deterministic via injected clock + seeded selection (HXC-1422).
    - `multiraft` (additive `lease.go`) — `LeaseTracker` CockroachDB-style leaseholder local reads (no Raft round-trip), follower routing via `RaftTransport.SendRead`, lease-expiry re-route against an injected clock (HXC-1389).

- Wave 69 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 68):
  - `stonith` — IPMI credential hardening: BMC password delivered via the `IPMI_PASSWORD` env + `-E` (never on the argv / process table) with redacted argv in error strings; tests assert the password appears in neither argv nor errors (HXC-908, closing a wave-67 follow-up).
  - `deviceplugin` — concurrency test driving `ApplyFingerprint`/`Allocate`/`Release` from 50+ goroutines under `-race`, proving the Registry never oversubscribes (mutex removal trips the race detector) (HXC-910, closing a wave-67 follow-up).
  - `internal/llm` — replaced the stub `Inference` with a real strategy-based router (latency/throughput/quality/cost) over the live model registry, excluding unhealthy/unloaded models with a typed no-eligible-model error and a real injectable `Backend` seam (HXC-1470).
  - `metrics` (additive `earningsmetrics.go`) — `EarningsMetrics` exposing `tao_earnings_total` / `graval_attestation_status` / `token_throughput` / `gpu_utilization` Prometheus-text series with deterministic (sorted) ordering + concurrent-safe setters (HXC-1483).
  - `internal/gpu` (additive `attesthook.go`) — GraVal attestation hook: `AllocateForChutes` refuses a GPU that has not passed attestation (`ErrAttestationRequired`), even on the idempotent re-allocation path; injectable `ChutesAttestor` seam (HXC-1448).
- Wave 70 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 71):
  - `provider/chutes` — OpenAI-compatible `/v1/chat/completions` provider (`cpk_` Bearer) with HTTP-429 retry+backoff; httptest proves retry-then-success, Bearer header, and typed errors (HXC-1510).
  - `pool` (additive `model.go`) — `GPUDevice`/`WorkloadSpec`/`GPUAllocation`/`PoolStatus` data model with snake_case JSON tags + round-trip/concurrency tests (HXC-1495).
  - `quantization` (additive `awq.go`) — AWQ 4-bit default serving format + `EstimateVRAM` (~25% of fp16) + budget-aware `SelectFormat` preferring AWQ when fp16 won't fit (HXC-1473).
  - `internal/gateway` (additive `inference.go`) — inference route forwarding to an injectable Chutes client; `TEERequired` => `X-E2EE-Enabled` routing marker; client failure => 502 (HXC-1469).
  - `internal/gpu` (additive `mig.go`) — MIG profile management: standardized vGPU tiers (1g.10gb/2g.20gb/3g.40gb/7g.80gb), in-memory partitioning with oversubscription rejection + fixed-at-creation immutability (HXC-1449).
- Wave 71 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 70; closes 4 prior follow-ups):
  - `multiraft` — RaftTransport async-delivery safety: inbox handler Steps under the shard mutex, outbound messages deferred + flushed after unlock; a new async-transport `-race` test proves a goroutine-delivering transport is race-free and still commits (HXC-909).
  - `kraft` — `CreateTopic` returns `ErrTopicExists` on a conflicting re-create (different partition count); identical re-create stays idempotent (HXC-912).
  - `porcupine` — WGL state-dedup memoization via `Model.Equal`: verdict-preserving pruning (memo fires on backtracking histories; identical PASS/FAIL + witness vs `Equal==nil`) (HXC-913).

        - `chutes` — `StreamChannel` honors ctx cancellation on a blocked read (prompt channel close, no goroutine leak) and closes the owned reader (HXC-911).
        - `marketplaceadapter` (additive `akash.go`) — Akash (Cosmos/AKT) adapter over an injectable client: reverse-auction pricing, SDL submission, provider-reputation gate, registered in the `Name()`-dispatch Registry (HXC-1458).
    - Wave 72 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
        - `provider/runpod` — RunPodProvider (implements `pool.GPUProvider`) with a warm-pool fast path over an injectable client (HXC-1515).
        - `provider/aws` — AWSProvider EC2 Spot adapter over an injectable EC2-client interface (no aws-sdk dep): GPU model→instance-type selection (p5/p4de/g5), Spot+tags, TOCTOU-safe capacity gate (HXC-1516).
        - `scheduler` (additive `tier_filter.go`) — GPUTier-aware filter predicate: hard-exclude disallowed tiers + preference scoring, composes with the existing count filter, fails closed on a nil predicate (HXC-1534).
        - `provider/chutes` — ctx-interruptible 429 backoff (select vs `ctx.Done()`) + honors `Retry-After` (delta-seconds/HTTP-date, `max(hint,backoff)`) (HXC-916, closing a wave-70 follow-up).
        - `grafanadash` (additive `tiercost_dash.go`) — Phase-8B tier-utilization + cost dashboard generator (panels target `gpu_tier_utilization/gpu_cost_per_hour`), valid + deterministic (HXC-1551).
    - Wave 73 (5 units, subagent-driven, 5 parallel streams in disjoint packages; impl→spec-review→quality-review + independent build/vet/-race/mutation gate):
        - `provider/ionet` — IONetProvider over the io.net REST API (implements `pool.GPUProvider`, injectable base URL + `*http.Client` + bearer): `DeployCluster` returns the backend-minted cluster run-UUID (parsed, never fabricated), `HealthCheck` derives healthy state, concurrency-safe reserve-under-lock capacity gate; closure proven against a `net/http/httptest` io.net backend (HXC-1514).
        - `gitops` — ArgoCD ApplicationSet federation client: matrix generator expands one Application per cell, `syncPolicy` prune+selfHeal, RollingSync canary→tier-2 (50%)→tier-1 (25%) ordering, drift/prune detection; logic proven vs an httptest ArgoCD, strictly-live UI capture honestly skipped (follow-up HXC-922) (HXC-1367).
        - `internal/federation` — Karmada PropagationPolicy + OverridePolicy engine (CRs modeled as Go structs → YAML, no client-go dep) with constraint-aware (data-locality/latency/cost/compliance) two-level cell selection and failover reselect; deterministic failover proven via in-process FakeHub, live Karmada <60s capture honestly skipped (follow-up HXC-924) (HXC-1364).
        - `health` (additive `miner.go`) — miner-api + GraVal bootstrap DaemonSet dependency checks wired into the health rollup, naming the specific failing check; healthy→unhealthy delta proven against an httptest miner-api (HXC-1484).
        - `discovery` (additive `mdns.go`) — mDNS/DNS-SD advertiser+browser on `_helix-cluster._tcp` (stdlib `net` + `x/net/dnsmessage`): TXT encode/decode + reject-invalid + in-process loopback discovery; real-

    - multicast two-node LAN capture sandbox-skipped (follow-up HXC-925); discovery-only (trust via SPIFFE) (HXC-1347).
    - Reconciled stale P0 `HXC-904` Phase-4 Build Service: delivered podman-based multi-arch container build orchestration + artifact cache (`internal/build/pkg/build/cmd/helix-build`, green) marked Completed; Bazel REAPI interop split to follow-up HXC-921.
- **Security fix (HIGH):** closed a TOCTOU replay-bypass in `pkg/gpuattest.Verify` — the nonce check and consume are now atomic, with a concurrent-replay `-race` regression test.
- Every foundation package ships with tests that fail under mutation of the logic they cover (CLAUDE-1 enforcement) and are gated by whole-tree `build`/`vet` + `-race`.
- Initial project scaffold with Go workspace.
- 29 submodules integrated at project root.
- Directory structure: `cmd/`, `pkg/`, `internal/`, `api/`, `web/`, `scripts/`, `deploy/`, `test/`.

Changed

- **Governance:** CLAUDE-3 / AGENT-2 / QWEN-2 documentation continuous-sync guarantee added (restates and cites Constitution §11.4.106 docs-chain); `.docs_chain/contexts/tracked_docs.yaml` extended to enforce README, CLAUDE/AGENTS/QWEN, CHANGELOG, FOUNDATION_PACKAGES, and CODEGRAPH export-sync.

[0.1.0-dev] - 2026-05-30

Added

- Phase 0 bootstrap: submodule cloning and workspace initialization.